Advanced Fixed Income and Credit

Group Assignment

Part I: Deutsche Bank — Finding Relative-Value Trades (HBS 205-059)
Part II: Python Labs — US Treasury Yield Curve Reaction to Trump Tariff Threats

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October 27, 2026

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1 Part I — The Deutsche Bank Case Study

Setup and conventions

Settlement date: **August 15, 2003**. Exhibit 1 lists 52 Treasury coupon bonds whose maturities fall on successive semi-annual dates (Feb 15 / Aug 15) from 2004 to 2029 with no gaps, i.e. exactly one bond at each of the nodes $t = 0.5, 1.0, \ldots, 26.0$ years. This allows the classic *successive substitution* bootstrap: because each bond introduces exactly one new unknown discount factor, we solve sequentially from the shortest to the longest maturity.

Compounding convention. Exhibit 4's footnote states explicitly that Deutsche Bank's own model yields are *Bond-Equivalent Yields* (BEY), i.e. twice the semi-annual compounded yield. Since U.S. Treasuries pay semi-annual coupons, we derive our own zero-coupon yields in the identical BEY convention throughout, so that every comparison against the DB model in Q1.2 is apples-to-apples (a continuously-compounded convention would have introduced a small but systematic bias in one direction).

$$\text{BEY} = 2\left(D^{-1/k} - 1\right), \qquad D = \text{discount factor to period } k \text{ (6-month steps)}.$$

Q1.1 — Bootstrapping the discount curve

Methodology. For bond k (maturing at period k, coupon c per annum, price P_k):

$$P_k = \frac{c}{2}\sum_{j=1}^{k-1} D_j + \left(\frac{c}{2} + 100\right) D_k \quad \Longrightarrow \quad D_k = \frac{P_k - \frac{c}{2}\sum_{j=1}^{k-1} D_j}{\frac{c}{2} + 100}.$$

Solved in order of increasing maturity, using only already-known D_j, $j < k$.

Validation. Our bootstrapped curve matches Deutsche Bank's own Exhibit 4 model yields *exactly* at 2Y (1.873%) and 10Y (4.711%) — precisely the maturities the case identifies as DB's own calibration points for its three-factor model. This is strong internal validation that both the bootstrap algorithm and the BEY convention choice are correct.

Maturity (y)	Discount Factor	Zero Yield (BEY)
0.5	0.99561	0.882%
1.0	0.98818	1.193%
2.0	0.96341	1.873%
5.0	0.84156	3.480%
10.0	0.62775	4.711%
15.0	0.42935	5.717%
20.0	0.30947	5.951%
25.0	0.23299	5.913%
26.0	0.22024	5.905%

Table 1: Bootstrapped zero-coupon curve, selected maturities (full 52-point curve in `q1_bootstrapped_curve.csv`).

The curve shape is the standard concave, upward-sloping Treasury curve of the period, flattening from about 15Y onward and showing a very slight downward drift between 20Y and 26Y.

Q1.2 — Comparison with Deutsche Bank's model, trading signals

Logic. Yield and price move in opposite directions. If our bootstrapped (market-implied) yield is *higher* than DB's model yield, the bond's market price is below the model's "fair" price ⇒ the bond is **cheap** ⇒ **BUY**. If the market yield is *lower*, the bond is **rich** ⇒ **SELL**.

Pattern. The belly of the curve (3–9 years) is uniformly *cheap* relative to the DB model, by 1–6 bps, suggesting a BUY candidate basket in that region. The very short end (1Y) and the long

Maturity (y)	DB Model Yield	Market Yield (Q1.1)	Diff (bps)	Signal
1	1.2443%	1.1930%	−5.13	SELL (rich)
2	1.8727%	1.8727%	+0.00	fair (DB calibration point)
3	2.4110%	2.4711%	+6.01	**BUY (cheap)**
4	2.9665%	3.0084%	+4.19	**BUY (cheap)**
5	3.4454%	3.4799%	+3.45	**BUY (cheap)**
6	3.8557%	3.8817%	+2.60	**BUY (cheap)**
7	4.1996%	4.2203%	+2.07	**BUY (cheap)**
8	4.4677%	4.4865%	+1.88	**BUY (cheap)**
9	4.6528%	4.6640%	+1.12	**BUY (cheap)**
10	4.7107%	4.7107%	−0.00	fair (DB calibration point)
15	5.7160%	5.7167%	+0.07	fair
20	5.9517%	5.9513%	−0.04	fair
25	5.9315%	5.9128%	−1.87	SELL (rich)

Table 2: Market vs. DB model comparison and trading signals.

end (25Y) look *rich*. The 2Y, 10Y, 15Y and 20Y points are essentially fair by construction or coincidence (2Y and 10Y are DB's own calibration anchors).

Q1.3 — Is this risk-free arbitrage?

No. Although the strategy is directional and model-based, it is **not** a riskless arbitrage, for several reasons:

- **Model risk.** The "mispricing" is only a mispricing relative to Deutsche Bank's own three-factor curve model. If the model is mis-specified (wrong number of factors, wrong functional form, stale calibration), the signal is simply wrong — there is no model-independent proof that the belly of the curve is "truly" cheap.
- **Convergence/timing risk.** Even if the model is correctly specified on average, there is no guarantee *when*, or even *whether*, market yields converge back toward the model yields within the trader's holding horizon. The position carries interim mark-to-market risk.
- **No riskless, capital-structure-free payoff.** A true arbitrage requires a replicating, self-financing portfolio with a knowable, riskless terminal payoff (e.g. a textbook cash-and-carry). Here the "trade" is an outright directional bet on convergence, funded at some financing rate, with basis risk between the position and any hedge.
- **Curve risk while waiting.** Between trade inception and convergence, the *whole* curve can move (parallel shifts, twists), and unless the position is duration/key-rate hedged, those moves swamp the small model-implied edge of a few basis points.

In short, this is a **relative-value** trade: a model-informed bet with positive expected value under the trader's own beliefs about the model's validity, not a riskless arbitrage in the classical sense.

Q2.1 — Cubic spline of the zero-coupon curve

Methodology. Natural cubic spline (zero second derivative at the boundary knots) through the five knots specified (2, 5, 10, 15, 20 years), fit directly on the Q1.1 bootstrapped BEY yields. Outside $[2, 20]$y the spline is extrapolated using its boundary slope.

Fit quality. Within the knot range $[2, 20]$y: mean residual $= -5.50$ bps, std $= 8.28$ bps, max $|\text{residual}| = 29.91$ bps. The residuals are *not* randomly scattered: the largest errors cluster sharply between 7.5 and 13 years (up to -30 bps at 11.5Y), exactly the maturity window in which Exhibit 1 contains several legacy, high-coupon bonds issued in the early 1980s (coupons of 13.25%, 12.5%,

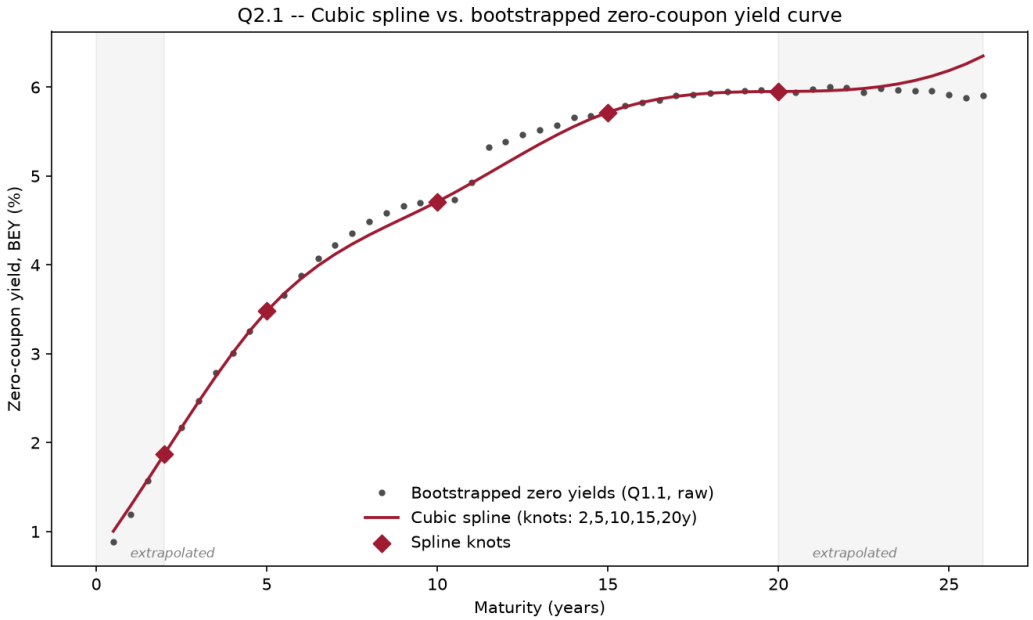


Figure 1: Cubic spline (red) vs. raw bootstrapped zero yields (grey dots). Shaded regions are extrapolated.

11.25%, 10.625%). These bonds plausibly carry idiosyncratic pricing noise (age, liquidity, possible embedded optionality/callability common to that vintage), which the raw bootstrap faithfully reproduces pointwise but which the smooth spline, by construction, averages away.

Q2.2 — Does using the spline change the Q1.2 conclusions?

We repeat the Q1.2 comparison using the **spline-implied** yield (instead of the raw bootstrap) at every maturity used in the original table, since the spline value differs from the raw bootstrap except exactly at the five knots (2,5,10,15,20).

Maturity (y)	DB Model	Spline Yield	Diff (bps)	Spline Signal	*Changed vs. Q1.2?*
1	1.2443%	1.2846%	+4.03	BUY	**YES** (was SELL)
2	1.8727%	1.8727%	+0.02	fair	no (knot)
3	2.4110%	2.4608%	+4.98	BUY	no
4	2.9665%	3.0096%	+4.31	BUY	no
5	3.4454%	3.4799%	+3.45	BUY	no (knot)
6	3.8557%	3.8439%	−1.18	SELL	**YES** (was BUY)
7	4.1996%	4.1195%	−8.01	SELL	**YES** (was BUY)
8	4.4677%	4.3363%	−13.14	SELL	**YES** (was BUY)
9	4.6528%	4.5235%	−12.93	SELL	**YES** (was BUY)
10	4.7107%	4.7107%	−0.00	fair	no (knot)
15	5.7160%	5.7167%	+0.07	fair	no (knot)
20	5.9517%	5.9513%	−0.04	fair	no (knot)
25	5.9315%	6.1859%	+25.44	BUY	**YES** (was SELL)

Table 3: Q1.2 signals recomputed off the Q2.1 spline curve.

This matters a great deal. Six of the thirteen signals (1, 6, 7, 8, 9, 25 years) *flip sign* once the spline replaces the raw bootstrap. Two distinct mechanisms are at work:

1. **Extrapolation risk (1Y and 25Y).** These two points lie *outside* the spline's knot range $[2, 20]$y. The natural spline's boundary-slope extrapolation is a pure modelling artifact with no

claim to describing the true curve shape beyond the data used to fit it, so any signal built on it is unreliable almost by definition.

2. **Belly-of-curve smoothing (6–9Y).** This is the more substantive finding. The original Q1.2 "cheap" signal in 6–9Y was driven, at least in part, by the *same* idiosyncratic legacy-bond noise identified in Q2.1's residual analysis. Once the spline smooths through that local bump, the belly of the curve actually looks *rich*, not cheap, relative to the DB model.

Conclusion. The trading conclusion from Q1.2 is *not robust* to the choice of curve-fitting methodology in exactly the region (6–9Y) where the raw bootstrap is most likely contaminated by non-fundamental, bond-specific noise. A prudent trader would treat the raw-bootstrap BUY signal in 6–9Y with real skepticism, and would want to understand *why* those bonds are mispriced (liquidity? embedded option? stale quote?) before trading on it, rather than assuming genuine relative value.

Q3.1 — Nelson-Siegel-Svensson fit

Methodology. Jointly estimating $(\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$ by unconstrained nonlinear least squares proved numerically fragile: this curve is concave with a single mild hump and no second hump, so β_0 (level) and β_2 (curvature) become highly collinear once τ_1 is pushed above ∼10 years (the curvature loading then barely decays within our 26-year range). Different, equally-well-fitting local optima emerged from different starting points.

We instead use the standard **Diebold–Li two-step procedure**: for each candidate (τ_1, τ_2) on a fine grid, the loadings become fixed basis functions of maturity, so the betas enter *linearly* and are solved by OLS — a well-posed, unique, convex problem for every candidate τ. We then pick the (τ_1, τ_2) minimizing total RSS.

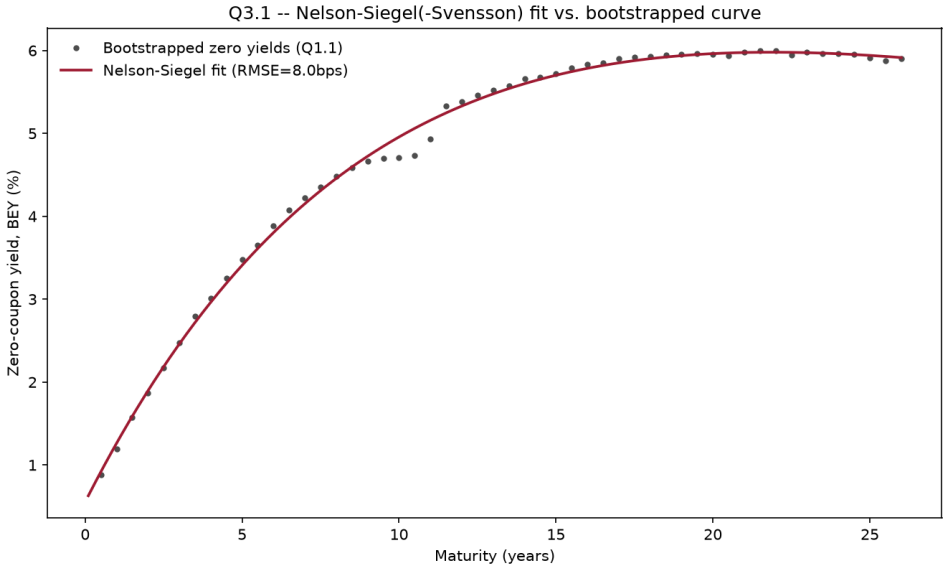


Figure 2: Nelson-Siegel fit vs. the Q1.1 bootstrapped curve.

Model selection. The plain 3-factor Nelson-Siegel achieves RMSE = 7.96 bps ($\beta_0 = 0.695\%$, $\beta_1 = -0.140\%$, $\beta_2 = 17.926\%$, $\tau_1 = 12.08$y). The full 4-factor Svensson only improves RMSE to 7.28 bps (an improvement of 0.68 bps), below our pre-set materiality threshold of 1.5 bps. We therefore adopt the **simpler, better-identified 3-factor Nelson-Siegel model**: the data show no evidence of a genuine second hump, so the extra parameter and its attendant identification problems are not worth the marginal fit improvement.

Parameter	Value
β_0 (level)	0.695%
β_1 (slope)	−0.140%
β_2 (curvature)	17.926%
τ_1	12.081 years
RMSE	7.96 bps

Table 4: Final adopted Nelson-Siegel parameters.

Q3.2 — Factor loadings and economic drivers

The contribution decomposition reveals a striking feature: the **curvature** factor ($\beta_2 = 17.9\%$) does almost all of the curve-shaping work, rising smoothly from 0 at $m = 0$ to its asymptote around 20 years, while the level contribution is small and essentially flat (0.695%) and the slope contribution is tiny and nearly flat (−0.140%). This is counter-intuitive at first glance — in textbook Nelson-Siegel intuition, "level" should dominate a curve of this overall magnitude — but it is a direct, mechanical consequence of τ_1 being large (12.08 years) relative to our 26-year maturity range: the curvature loading function $\left[\frac{1-e^{-m/\tau_1}}{m/\tau_1} - e^{-m/\tau_1}\right]$ simply has not decayed back to zero by $m = 26$, so it is statistically *behaving like* a second, slowly-varying level/slope factor over the range of our data, and the OLS step naturally loads most of the explanatory power onto it rather than onto β_0 precisely because β_0's loading is flat and uninformative about curve *shape*, whereas the curvature loading can trace the whole concave shape on its own.

Economically, this echoes Deutsche Bank's own Exhibit 3 three-factor model description: a persistent, long-horizon "long-rate"/level-type factor that evolves slowly is naturally captured by a Nelson-Siegel factor whose decay parameter is large relative to the fitting window, rather than by the model's nominal level factor, which by construction carries no shape information at all. In other words, when the sample's maturity range is not much longer than τ_1, Nelson-Siegel's "curvature" factor and a slow-moving macro "long rate" factor become difficult to tell apart — exactly the level/curvature identifiability issue we had already flagged during estimation (Q3.1).

Q4 — Answer summary

Q4.1. A +50bps bump of the 10Y knot with the natural spline re-fitted raises the 10Y zero yield by exactly 50bps and leaves the other four knots unchanged, but the shock is *not* local: the curve ripples (+22.1bps at 7Y, −8.4bps at 17Y, −11.5bps at 26Y).

Q4.2. KRD_{10Y} = 2.21 years (ΔP = −\$10,813.14, i.e. −1.08% of NAV; DV01 = \$221.0 central, \$216.3 one-sided).

Q4.3. The 52 individual zero-yield KRDs sum to 7.981 years, exactly the portfolio's effective duration; the aggregated key-rate profile is 0.37/1.54/2.21/0.81/3.06 years at 2/5/10/15/20Y.

Q4.1 — Shocking the 10Y key rate

Methodology. The Q2.1 curve is a natural cubic spline through the bootstrapped zero yields at the knots 2, 5, 10, 15 and 20y, which are exactly the assignment's key rates. We therefore implement "shock the 10-year rate by 50bps" literally on that model: we add +50bps to the 10Y knot value, *re-fit* the natural spline, and define the curve shock as

$$\Delta y(t) = S_{\text{bumped}}(t) - S_{\text{base}}(t), \qquad t = 0.5, 1.0, \ldots, 26\text{y}.$$

Because spline interpolation is linear in the knot values, $\Delta y(t)$ does not depend on the level of the curve, and the five key-rate shocks sum *exactly* to a +50bps parallel shift at every maturity (verified in the code). For comparison, we also run the textbook Ho (1992) triangular ("tent") shock: full size at its own knot, decaying linearly to zero at the adjacent knots, and held flat beyond the outer knots.

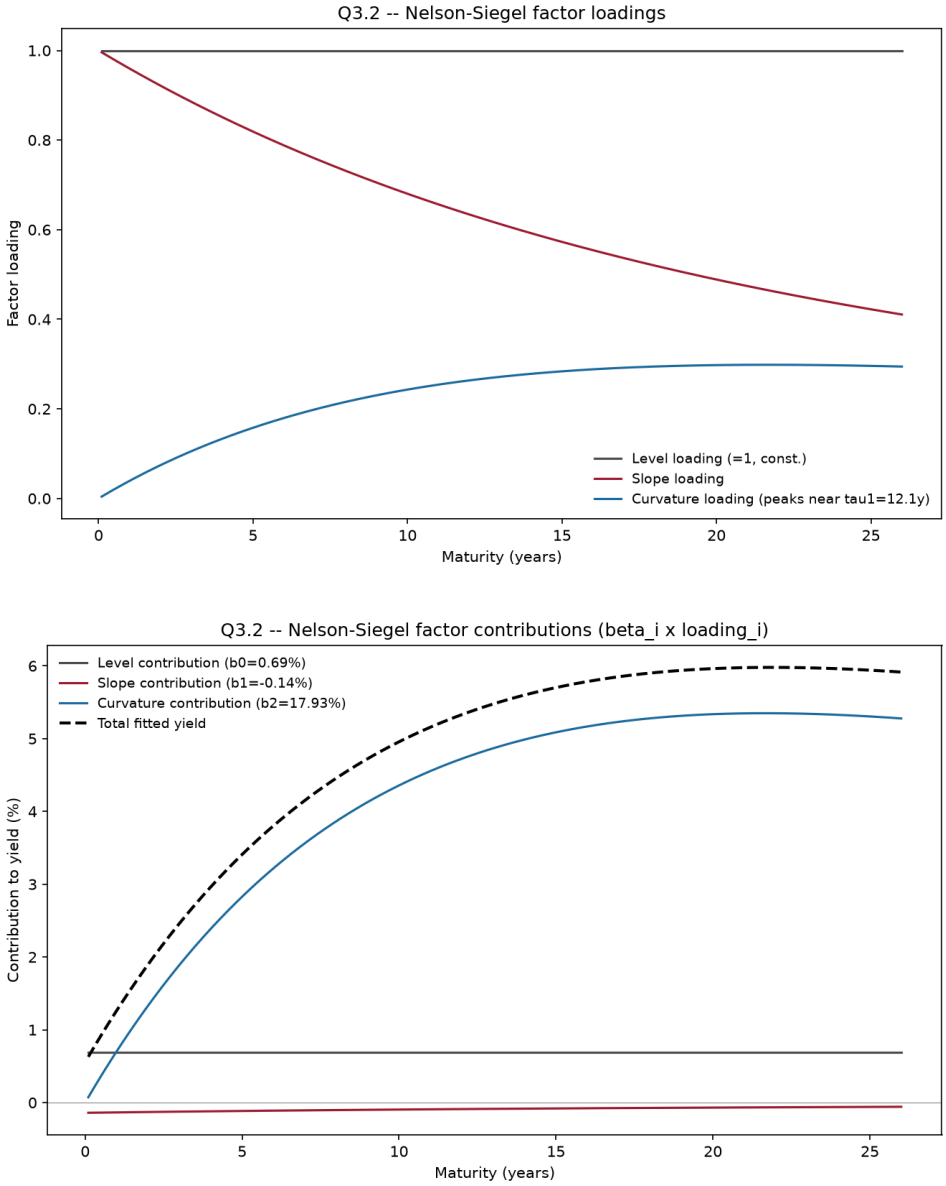


Figure 3: Top: factor loadings. Bottom: factor contributions ($\beta_i \times$ loading$_i$).

Impact on the curve. The shock is +50bps at 10Y and, by construction, zero at the other knots (2, 5, 15, 20y). However, unlike the tent, the spline response is **not local**:

- Between 5Y and 15Y it is *wider* than the tent: +22.1bps at 7y and +37.0bps at 12y, versus +20 and +30bps for the linear ramp.
- Outside $[5, 15]$y the cubic *ripples* with alternating signs: about −3 to −4bps between 3 and 4y, −8.4bps at 17y, and +8.4bps at 23y.
- In the **extrapolated** region beyond 20y the spline continues the last cubic polynomial — the `CubicSpline` implementation extends the end cubic, it does not hold the curve flat — so the ripple grows to −11.5bps at 26y. (A flat extension would put the 26Y yield at 5.95% instead of 6.35% and would set this last contribution to zero; we keep the model-consistent extension and quantify the sensitivity in Q4.3.) The right panel shows that this effect is much stronger for the outer knots: a 20Y bump moves the 26y yield by about +103bps, a 15Y bump moves it by about −44bps, and a 2Y bump moves the 0.5y yield by +79bps.

In short, a key-rate shock is only as local as the curve model it is applied to. The tent is local by design, while a cubic spline spreads a single-node shock across the whole curve, most strongly where the curve is extrapolated rather than interpolated.

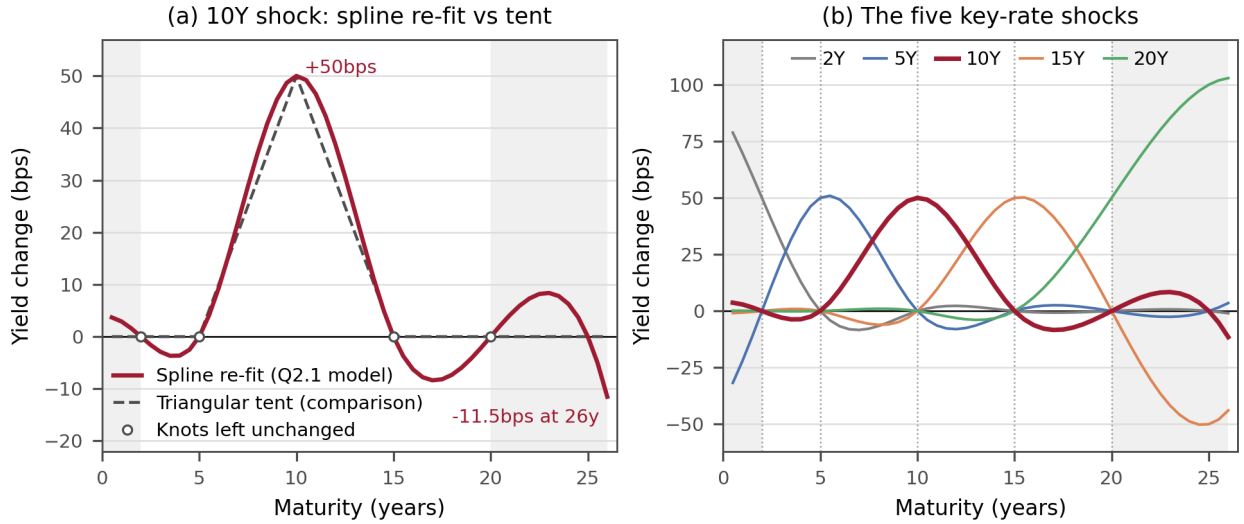


Figure 4: Left: the curve shock $\Delta y(t)$ from re-fitting the spline (red) vs. the triangular tent (dashed); hollow markers are the four knots left unchanged. Right: the five spline key-rate shocks. Shaded areas: extrapolated maturities (< 2y, > 20y).

Maturity (y)	2	3	5	7	10	12	15	17	20	23	26
Δy, spline re-fit (bps)	0.0	−2.9	0.0	+22.1	+50.0	+37.0	0.0	−8.4	0.0	+8.4	−11.5
Δy, tent (bps)	0.0	0.0	0.0	+20.0	+50.0	+30.0	0.0	0.0	0.0	0.0	0.0

Table 5: The +50bps 10Y shock at selected maturities: global spline re-fit vs. the local triangular tent. Δy is zero at 5Y and 15Y because the tent is a straight line between the knots; it is zero at 2, 5, 15 and 20Y for the re-fit because those knots are not moved.

Q4.2 — Portfolio KRD to the 10Y shock

Portfolio construction (assumption). "Equally invested" is read as an equal *dollar market value*: \$1,000,000/52 ≈ \$19,230.77 in each bond, i.e. a face amount of (dollar amount/price) × 100 per bond. The alternative reading, an equal *face* amount (\$19,230.77 of face in each bond), would give a \$1.19m market value and $\text{KRD}_{10Y} = 2.31$y: equal face overweights the cheap short bonds and underweights the expensive long ones, so it raises duration. We use equal market value, the standard reading of "equally invested", and report the alternative in the robustness table.

Approach.

1. Build the 52 × 52 cash-flow matrix (semi-annual coupons plus principal) and aggregate it into the portfolio cash flow at each semi-annual date.
2. *Base valuation* uses the Q1.1 bootstrapped zero curve (BEY, semi-annual compounding). It reprices every bond exactly, so the base NAV is exactly \$1,000,000. Pricing off the spline itself would give \$1,002,389, and the \$2,389 gap (the Q2.2 fitting residual) would contaminate the risk measure.
3. *Shocked valuation*: add the spline-implied $\Delta y(t)$ from Q4.1 to the bootstrapped yields and reprice. The shape of the shock therefore comes from the Q2.1 model, while the base level matches market prices.
4. Report the result as (i) the dollar P&L of the +50bps shock; (ii) the KRD in years, using a central difference to strip out convexity, $\text{KRD}_k = -\dfrac{P(+h) - P(-h)}{2\,P_0\,h}$ with $h = 50$bps; and (iii) the DV01 (\$ per 1bp), which is the quantity used to size hedges.

Result.

$\Delta P_{10Y} = \$989{,}186.86 - \$1{,}000{,}000.00 = -\$10{,}813.14$ (−1.08% of NAV), $\text{KRD}_{10Y} = 2.21$ years, DV01_1

The sign is as expected: a long bond portfolio loses value when yields rise. The \$221.0 is a *central-difference* DV01, DV01 $= -(P(+h)-P(-h))/2\times10^{-4}$; the *one-sided* +50bps P&L implies \$216.26 per bp, so $\Delta P \neq -50 \times$ DV01. The \$4.8 per bp gap is the convexity term, which a 50bp move is large enough to reveal; we report the central figure because it is the one used to size the hedge in Q4.3.

What actually moves the number (robustness). The KRD is essentially invariant to the pricing engine and materially sensitive to the *shock design*, which is where most of the dispersion between reasonable implementations comes from:

Variant	ΔP for +50bps	KRD_{10Y} (yrs)
Spline re-fit, bootstrapped base (**reported**)	−\$10,813	2.210
Spline re-fit, spline-priced base	−\$10,954	2.234
Triangular tent, bootstrapped base	−\$10,123	2.061
Triangular tent, spline-priced base	−\$10,232	2.078
Single-point 10Y bump (10Y zero yield only)	−\$1,169	0.240
Equal-face portfolio, bootstrapped base	−\$13,445	2.307

Every defensible convention puts the 10Y exposure between 2.0 and 2.4 years; the spread is methodological, not arithmetic. We report 2.21y because (i) Q4.1 asks for the impact on the curve *estimated in Q2.1*, which is the re-fitted spline, and (ii) pricing the base off the bootstrap keeps P_0 exactly at the \$1m actually invested, so the −\$2,389 Q2.2 fitting residual cannot leak into the risk measure. The 0.24y row shows the opposite extreme: applying the shock to the single 10Y zero yield only measures the partial duration of one cash-flow date, not a key-rate duration.

Q4.3 — Full key-rate duration profile

(a) Each of the 52 zero-coupon yields individually (the literal reading). "All maturities" is read as applying the same +50bps shock to each semi-annual zero yield (0.5y to 26y) *on its own*, leaving the other 51 unchanged. This version needs no curve model: only the discount factor of that single date moves, so to first order the KRD at date t is $t \times \text{PV}(\text{CF}_t)/\text{NAV}$ (i.e. $[t/(1+y_t/2)] \times \text{PV}(\text{CF}_t)/\text{NAV}$ in BEY terms).

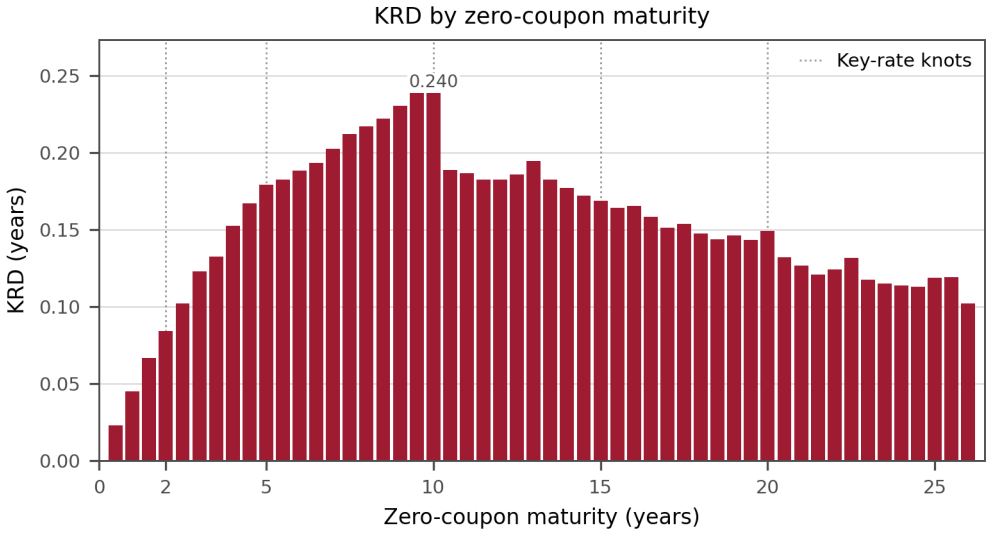


Figure 5: KRD to each individual zero-coupon yield (+50bps). The 52 bars sum to 7.981 years, the portfolio's effective duration.

Additivity check. The 52 KRDs sum to 7.9809 years, exactly the effective duration of a ±50bps parallel shift: a parallel shift is the sum of the 52 single-date shocks, so this check needs no curve model and is free of cross terms.

Shape of the profile. The exposure at date t is roughly $t\times$(PV of the portfolio cash flow at t)/NAV. It rises with t to a peak of 0.24y at 10y, then *drops abruptly* at 10.5y. Bonds maturing from Feb-2014 onwards carry high coupons (5.25–13.25%) and mostly trade far above par (up to 174), so an equal *dollar* allocation buys less face — hence less principal — at those dates. Beyond 10.5y the exposure declines slowly as discounting erodes the PV of distant cash flows.

(b) The five key rates (aggregated view). We apply the same +50bps re-fit shock to each knot in turn.

	Spline re-fit (Q2.1 model)			Tent (comparison)	
Key rate	P&L ($)	KRD (yrs)	DV01 ($)	P&L ($)	KRD (yrs)
2Y	−1,820	0.367	36.7	−2,640	0.531
5Y	−7,594	1.541	154.1	−7,156	1.446
10Y	−10,813	2.210	221.0	−10,123	2.061
15Y	−3,496	0.810	81.0	−8,236	1.690
20Y	−13,980	3.064	306.4	−10,717	2.252
Sum	−37,704	7.992	799.2	−38,872	7.979

Table 6: Key-rate exposures of the $1M equally-weighted portfolio (+50bps shock at each knot). The parallel-shift effective duration is 7.981 years. KRD and DV01 are central differences; the P&L column is one-sided, so P&L $\neq -50\times$DV01.

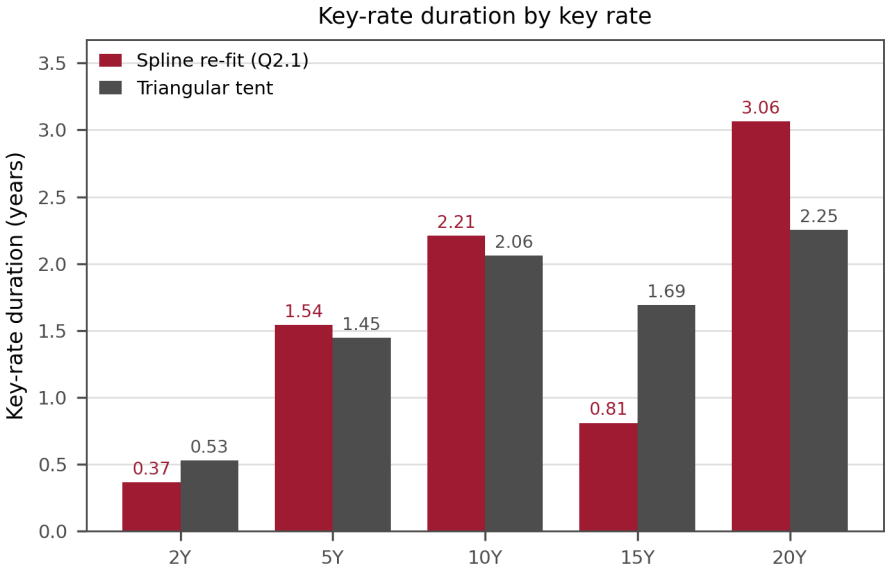


Figure 6: Key-rate duration profile (years): spline re-fit vs. triangular tent.

Consistency check. Both families of shocks sum to a parallel shift, so the KRDs should add up to the effective duration. They do: 7.992y (spline) and 7.979y (tent), versus 7.981y for a ±50bps parallel shift. Summing the *one-sided* dollar P&Ls gives −$37,704 versus −$38,594 for the parallel shock. The gap comes from convexity and cross terms, which are largest for the spline because its extrapolated shocks reach about 100bps. This is why we report central-difference KRDs.

Robustness: the 15Y/20Y split is an extrapolation effect. The Q2.1 spline is pinned by data only up to 20Y; beyond that, the end cubic amplifies the effect of a 50bps bump at 20Y to +103bps at 26Y, and the effect of a 50bps bump at 15Y to −50bps at 24.5Y. That is what moves duration between the two buckets:

	2Y	5Y	10Y	15Y	20Y	Sum
End cubic (reported)	0.367	1.541	2.210	0.810	3.064	7.992
Held flat beyond 20Y	0.180	1.619	2.109	1.850	0.652	6.411

Holding the curve flat would move the split to 1.85y (15Y) and 0.65y (20Y), but the five KRDs would then sum to 6.41y instead of the portfolio's 7.98y effective duration, because the shocks would no longer sum to a parallel shift: the cash flows beyond 20Y, about 1.57y of duration, would carry no key-rate exposure at all. We keep the model-consistent extension — it is the only one under which the key-rate decomposition is complete — and present the 15Y/20Y split as model risk rather than as a tradable exposure.

Interpretation.

- All exposures are positive durations (negative P&L), as expected for a long-only portfolio.
- With the tent, the profile is smooth (0.53 / 1.45 / 2.06 / 1.69 / 2.25y) and follows the cash-flow intuition; the 20Y bucket is the largest because it also absorbs every maturity from 20 to 26y.
- With the spline re-fit, the ripples of Q4.1 redistribute risk. The 2Y KRD falls to 0.37y (vs. 0.53y): the 2Y bump pushes 5–10y yields *down* by up to 8.4bps, worth +\$1,063 on the 5–10y cash flows, which more than offsets the extra −\$219 caused by its larger short-end shock (+79bps at 0.5y vs. +50bps for the tent). The 15Y/20Y split moves as explained above. The total is unchanged, which is the point.
- **Practical implication.** The choice of curve model is itself a source of hedging risk. A hedge sized on spline KRDs would put far more weight on 20Y and far less on 15Y than a hedge sized on tent KRDs, even though both imply the same total duration.

Illustrative KRD hedge (beyond the question). Neutralising the five spline key-rate DV01s with the five bonds maturing exactly at the key dates requires short \$93.6k (Aug-2005), short \$199.5k (Aug-2008), short \$211.5k (Aug-2013), *long* \$19.4k (Aug-2018) and short \$445.6k (Aug-2023), for a hedge market value of −\$968k (the funding leg is not modelled). The +50bps parallel P&L falls from −\$38,594 to +\$57; the individual key-rate residuals are at most \$606 (20Y), from second-order convexity.

2 Part II — Python Labs: US Treasury Yield Curve Reaction to Trump Tariff Threats

Data, event set and conventions

Data. Daily US Treasury constant-maturity yields (FRED `DGS3MO` to `DGS30`), downloaded with the course helper `fred.py`: 2015-01-02 to 2025-08-01, i.e. 2,647 trading days (Q1). **Code:** script `06_event_study.py` in the `ga_project` folder. One run reproduces every number, table and figure of this part (outputs prefixed `p1_`). The original notebook is in `ga_project/python_labs`.

Events. The notebook's `trump_actions` and `country_responses` dictionaries give 75 events (36 Trump actions, 39 foreign responses). Inside the sample they map to **58 distinct trading days** at horizon −30 (51 at +30): some events fall on the same day (a Trump action and a foreign response on 2018-07-06, 2018-08-23, 2018-09-24, 2019-08-23, 2019-09-01 and 2025-03-04, while the weekend events of 5 and 6 April 2025 both map to Monday 7 April), and the six events after 2025-08-01 are dropped because the FRED download stops on that date.

Methodology (as required). For each event day t and horizon $k \in [-30, +30]$ trading days we compute $y_{t+k} - y_t$, average it across events, and subtract the same-horizon average change over all non-event days (more than 30 trading days away from any event):

$$\mathrm{AC}(k) = \overline{y_{t+k} - y_t}^{\text{events}} - \overline{y_{t+k} - y_t}^{\text{non-events}}.$$

Level = 10Y, slope = 10Y−2Y, curvature = 2×5Y−2Y−10Y.

Reading the results (our convention). Because AC(k) is measured relative to day 0, a *positive* value at $k < 0$ means that yields were *higher* before the event, i.e. they *fell* into it. To avoid this sign trap, the tables report three windows, all as "abnormal yield change over the window" (positive = yields rose): pre $[-30, 0] = -\mathrm{AC}(-30)$, announcement $[-1, +1] = \mathrm{AC}(+1) - \mathrm{AC}(-1)$, post $[0, +30] = \mathrm{AC}(+30)$.

Assumptions. (A1) Business days = trading days in the FRED data; weekend/holiday events move to the next trading day. (A2) "Reaction of targeted countries" (Q1.3) is measured on the *US* curve on the dates of the foreign responses, since only US yields are provided. (A3) Q1 uses the notebook's event list and sample unchanged; all corrections are in Q2.1.

Q1.1 — Market reaction to Trump's announcements and actions

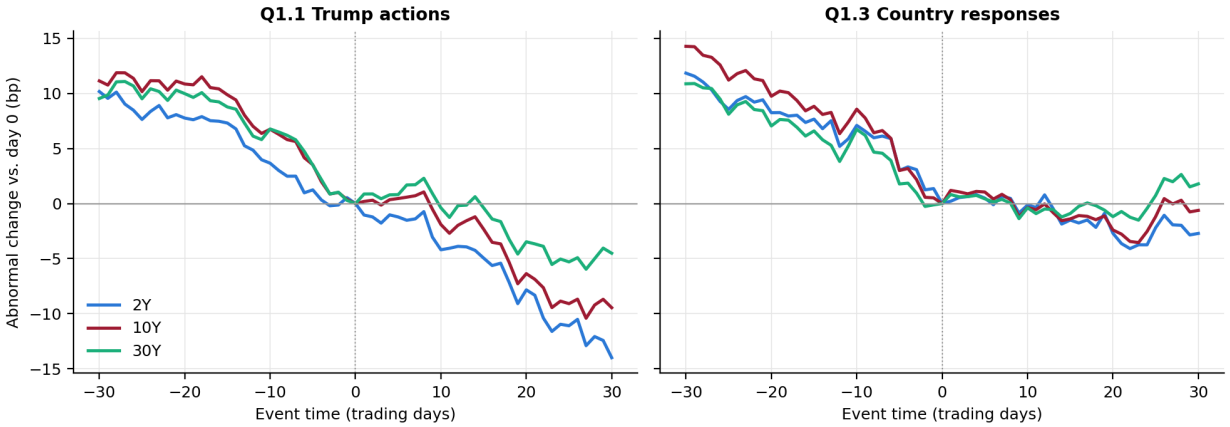


Figure 7: Abnormal change relative to the event day (bp), assignment methodology. Left: Trump actions (Q1.1). Right: foreign responses (Q1.3). A line that falls from left to right means yields fell through the window.

Discussion.

Window (bp)	3M	2Y	5Y	10Y	30Y
pre $[-30, 0]$	-6.4	-10.2	-11.6	-11.1	-9.5
announcement $[-1, +1]$	-0.5	-1.6	-0.8	-0.2	$+0.6$
post $[0, +30]$	-7.1	-14.0	-13.6	-9.5	-4.5

Table 7: Trump actions: abnormal yield change (32 event days at -30, 28 at $+30$).

- **No visible announcement effect.** Around the event itself the curve barely moves (-1.6bp on the 2Y, -0.2bp on the 10Y over $[-1, +1]$), small compared with a daily standard deviation of about 5bp.
- **Large drift on both sides.** Yields fall by about 10–12bp in the 30 days *before* the events and by a further 5–14bp *after* them. Taken at face value, the market reads tariffs as a negative growth shock: flight to quality into Treasuries and higher odds of Fed cuts, which is why the front end (2Y, -14.0bp post) falls the most.
- **But the shape is suspicious.** A genuine reaction to news should be a jump around day 0, not a smooth 60-day trend that starts a month before the announcement. The pattern is more consistent with events clustering in falling-rate periods (2019 easing cycle, 2025 cut expectations) than with a tariff effect. Q2.1 tests this directly.

Q1.2 — Level, slope and curvature

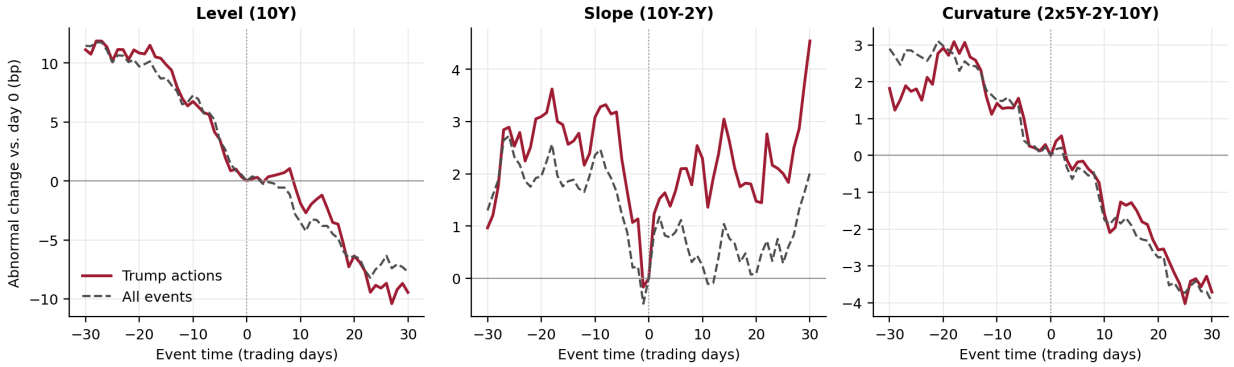


Figure 8: Abnormal change of level, slope and curvature (bp): Trump actions (red) vs. all 75 events (dashed).

	Trump actions			All events		
Window (bp)	Level	Slope	Curv.	Level	Slope	Curv.
pre $[-30, 0]$	-11.1	-1.0	-1.8	-11.5	-1.3	-2.9
announcement $[-1, +1]$	-0.2	$+1.4$	$+0.1$	-0.2	$+1.3$	-0.1
post $[0, +30]$	-9.5	$+4.5$	-3.7	-7.7	$+2.0$	-4.0

Table 8: Abnormal change of the three curve components.

Discussion.

- **Level** carries almost all of the movement and replicates Q1.1 (it is the 10Y by construction, a redundancy we address with a PCA level in Q2.1).
- **Slope.** After Trump actions the curve *steepens* ($+1.4$bp at the announcement, $+4.5$bp over 30 days) while the level falls: a *bull steepening*, i.e. the front end rallies more than the long end. This is the typical signature of markets pricing a more accommodative Fed: the 2Y is anchored to expected policy rates, while the 10Y also carries inflation risk from tariffs and a

term premium.

- **Curvature** falls (−3.7bp post): the 5Y belly rallies more than the average of the 2Y and 10Y wings, consistent with the market pricing cuts concentrated over the next 2–5 years rather than permanently lower rates.
- The pattern is the same for all events and slightly weaker, because foreign responses (Q1.3) contribute little post-event movement.

Q1.3 — Market reaction to the targeted countries' responses

Window (bp)	3M	2Y	5Y	10Y	30Y	Slope	Curv.
pre $[-30, 0]$	−2.6	−11.8	−15.3	−14.3	−10.9	−2.4	−4.4
announcement $[-1, +1]$	+0.1	−1.1	−0.2	+0.7	+0.9	+1.8	0.0
post $[0, +30]$	−3.2	−2.7	−3.7	−0.6	+1.8	+2.1	−4.1

Table 9: Foreign responses: abnormal yield change (34 event days at −30, 31 at +30).

Discussion.

- **Before** the responses, yields fall by 11–15bp, as before Trump actions. This is expected: responses come days after the US measure they answer, so the pre-window mostly contains the US action itself.
- **After** the responses, there is almost no further drift (−0.6bp on the 10Y, +1.8bp on the 30Y): retaliation seems largely anticipated once the US measure is known.
- The long end *rises* relative to the front end (slope +1.8bp at the announcement, +2.1bp post). Retaliation hurts US exporters (growth-negative) but also raises import prices and threatens foreign demand for Treasuries (inflation and term-premium positive); at the long end the second channel dominates.
- **Caveat.** Seven response days coincide with a Trump action on the same trading day, so the two effects cannot be separated there (fixed in Q2.1).

Q1.4 — First vs. second mandate

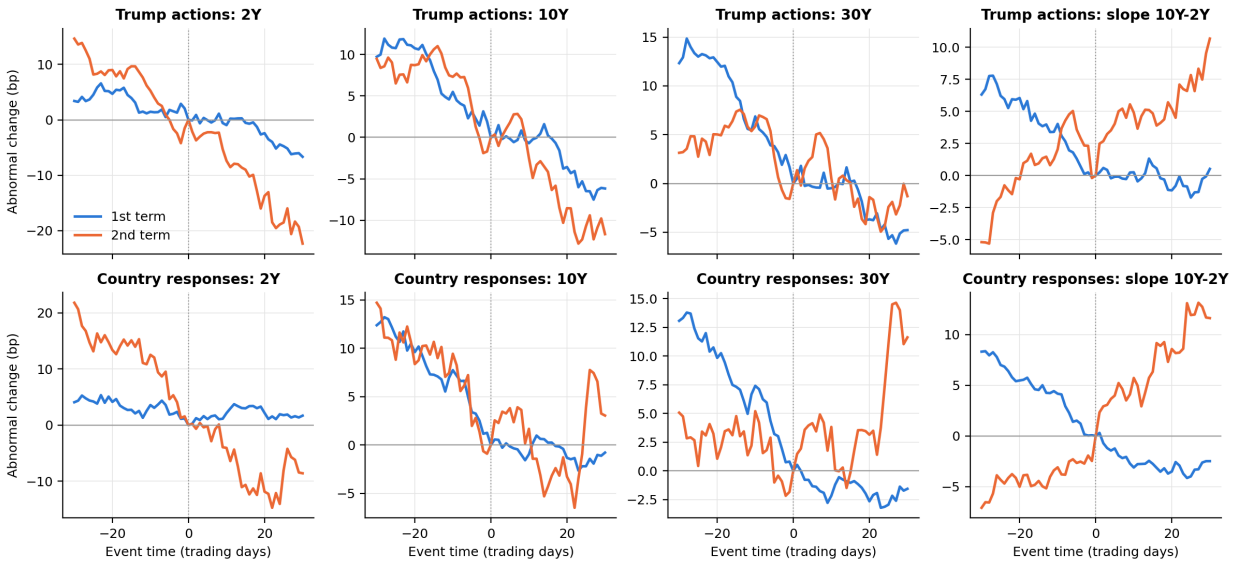


Figure 9: Abnormal change by mandate: first term (2017–20, blue) vs. second term (2025, orange). Top: Trump actions. Bottom: foreign responses.

Discussion. The reactions differ markedly.

		pre $[-30, 0]$ (bp)				post $[0, +30]$ (bp)			
Group	n	2Y	10Y	30Y	Slope	2Y	10Y	30Y	Slope
Trump actions, 1st term	18/18	−3.4	−9.7	−12.3	−6.3	−6.7	−6.2	−4.8	+0.5
Trump actions, 2nd term	14/10	−14.7	−9.5	−3.1	+5.2	−22.4	−11.7	−1.3	+10.7
Responses, 1st term	22/22	−4.1	−12.4	−13.1	−8.3	+1.7	−0.8	−1.6	−2.5
Responses, 2nd term	12/9	−21.8	−14.7	−5.1	+7.1	−8.6	+3.0	+11.6	+11.6

Table 10: Abnormal yield change by mandate; n = event days used at −30/+30.

- **First term (2018–19): a parallel, safe-haven rally.** 2Y, 10Y and 30Y fall by similar amounts and the slope barely moves. Tariffs were targeted (mainly China), inflation was low, and Treasuries behaved as the classic safe asset.
- **Second term (2025): a steepening.** After Trump actions the 2Y falls by 22bp while the 30Y is unchanged (slope +10.7bp); after foreign responses the 30Y even *rises* by 11.6bp. The front end prices Fed cuts against weaker growth, but the long end no longer rallies: universal tariffs are inflationary, US fiscal deficits were widening (One Big Beautiful Bill Act), and markets questioned foreign demand for Treasuries. The term premium absorbed the shock, and the safe-haven role of the long bond was weaker.
- **Reliability.** The second-term samples are small (10 and 9 events at +30) and concentrated in a few weeks of 2025, so these differences are indicative only. Q2.1 tests which of them survive.

Q2.1 — Limitations, improvements, and how they change Q1

(a) Limitations of the design

1. **Biased baseline (regime effect).** The "normal" change is estimated on non-event days, which fall mostly in 2015–16 and 2021–24, the Fed hiking cycle. The 10Y non-event baseline rises by +6.5bp per 30 days, while the average 30-day change was −3.6bp in 2017–20 and −5.8bp in 2025 (Table 11). Subtracting a rising baseline from falling-rate periods mechanically creates a negative "abnormal" change: a large part of the Q1.1 post-event drift is a regime effect, not a tariff effect.
2. **Overlapping windows.** Events come in bursts (e.g. 14 events between 2025-02-01 and 2025-04-10), so 30-day windows overlap and contain each other's events. Observations are not independent, and the long windows mix many news items.
3. **No statistical inference.** Averages are reported without standard errors; given a daily standard deviation of about 5bp, a few bp over 30 days is well within noise.
4. **Event-list quality.**
   - Two entries are silently lost to duplicated dictionary keys (2025-08-06 and 2025-03-04).
   - Some entries are not shocks: "no retaliation" announcements, a status quo, a US court ruling listed as a foreign response.
   - Escalations and de-escalations (Phase One deal, delays, exemptions) are averaged together, so they offset each other.
   - Major milestones are missing: the 90-day pause (2025-04-09), the US–China Geneva agreement (2025-05-12) and the US–EU deal (2025-07-27).
   - Some descriptions are inaccurate (145% was the US rate on China).
5. **Sample truncation.** The download stops on 2025-08-01: six August-2025 events are dropped and late-July windows are cut short.
6. **Timing.** Several announcements came after the market close ("Liberation Day" on 2025-04-02 was announced after 4pm ET) or on weekends, so day 0 is not the reaction day (Figure 11).
7. **Confounding news.** Some event days coincide with FOMC decisions; Trump actions and foreign responses share the same trading day seven times (eight in the extended sample).

8. **Measurement.** Level = 10Y duplicates Q1.1. Constant-maturity par yields are used rather than zero rates, and daily closes cannot isolate an announcement from other news on the same day.

Period	2015–16	2017–20 (1st term)	2021–24	2025 (2nd term)
Mean 30-day 10Y change (bp)	+3.3	−3.6	+9.7	−5.8

Table 11: Average 30-day change of the 10Y yield by period: the non-event baseline (+6.5bp) is dominated by the 2021–24 hiking cycle.

(b) Improvements implemented

All are in `06_event_study.py`, function `part_q2`.

1. **Cleaned and signed event list.**
   - Duplicated keys merged; six non-events removed (list in the code).
   - Each event signed as escalation (+1) or de-escalation (−1).
   - Three missing de-escalation milestones added.
   - Result: 72 events, of which 63 escalations and 9 de-escalations.
2. **Extended sample** to 2025-10-31, so that every event has a full window.
3. **Short windows** $[-1,+1]$ and $[0,+5]$ to capture the announcement effect. $[-1,+1]$ also covers after-close announcements, since the reaction then occurs on day +1.
4. **Local baseline.** Expected change = average daily change in the *same mandate* on clean days (no event within ±5 days), times the window length. This is a constant-mean model by regime, instead of a 2015–2025 average.
5. **Placebo inference.** For each group we draw 5,000 sets of pseudo-event dates (same number per mandate, from clean days of that mandate). The two-sided p-value is the share of placebo averages at least as extreme as the actual one.
6. **De-clustering.** The 30-day window uses only the first event of each cluster (no other event in the previous 5 trading days).
7. **Robustness.** We exclude events with a same-day foreign response, events within ±1 day of an FOMC decision, and replace the 10Y by a PCA level (first principal component of daily changes).

	$[-1,+1]$				$[0,+5]$			$[0,+30]$, cluster heads		
Group	2Y	10Y	30Y	Slope	10Y	30Y	Slope	2Y	10Y	n
Trump escalations	−0.1	+0.1	+1.1	+0.2	+0.1	+0.9	+1.6**	+2.9	+3.0	31/18
1st term	−2.1	−2.2	−1.7	0.0	−0.7	−0.8	−0.7	+9.0	+7.2	16/12
2nd term	+2.1	+2.6	+4.1**	+0.5	+0.9*	+2.6**	+4.0**	−9.3	−5.5	15/6
Foreign escalations	+0.8	+2.5*	+2.9**	+1.6**	+3.0**	+3.2**	+1.7**	+1.4	+4.8	32/10
1st term	+0.4	+0.4	+0.6	0.0	+0.6	−0.7	−1.7	+10.7	+7.7	20/7
2nd term	+1.6	+5.9**	+6.9**	+4.3**	+7.1**	+9.5**	+7.2**	−20.3**	−2.0	12/3
De-escalations (all)	+5.1**	+4.8*	+4.6*	−0.3	−0.4	+1.4	+2.2*	−26.7**	−24.5**	9/4

Table 12: Improved design: abnormal change in bp (positive = yields up), regime-specific baseline, extended sample, cleaned events. Placebo p-values: $^{*}p < 0.10$, $^{**}p < 0.05$. n = events in the short / 30-day windows.

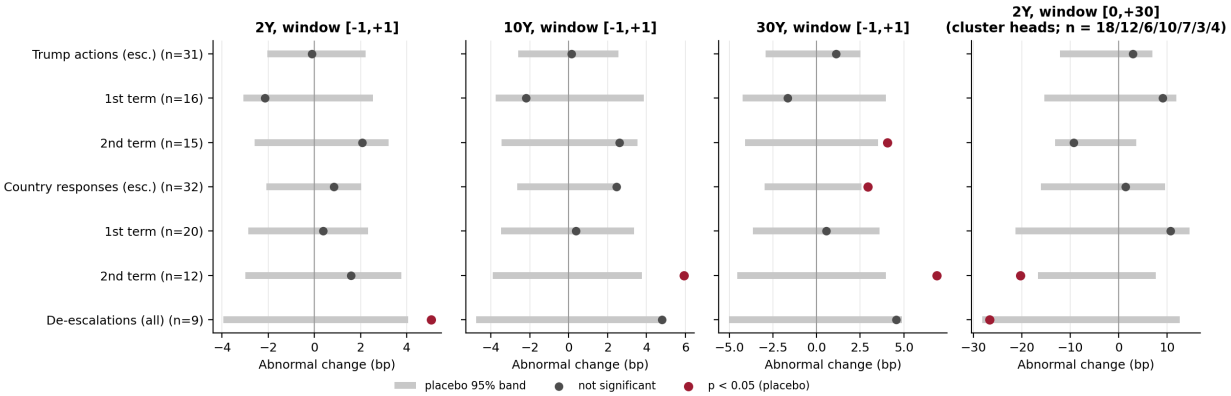


Figure 10: Improved estimates (dots) against the 95% range of the placebo distribution (grey bars). Red dots are significant at 5%.

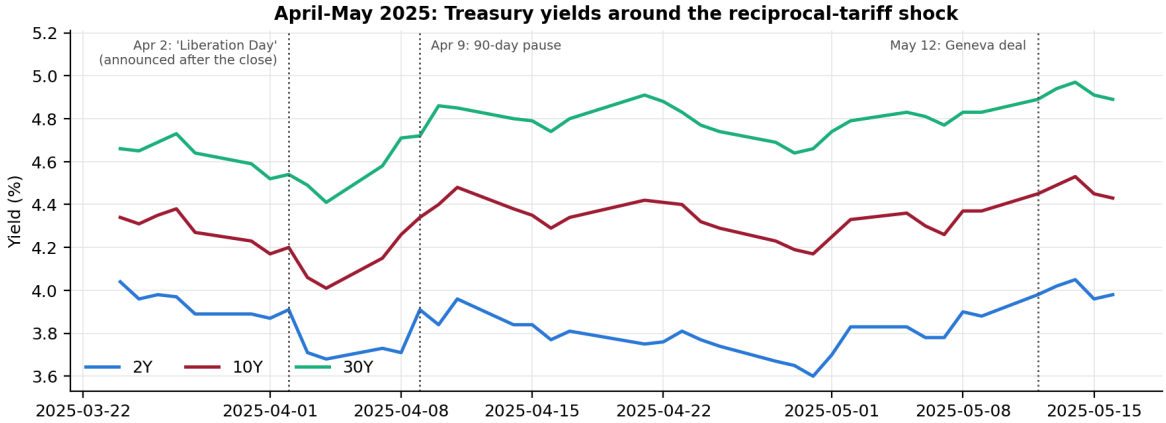


Figure 11: Daily yields around "Liberation Day". The announcement came after the close on 2 April, so the reaction is on 3 April (−20bp 2Y, −14bp 10Y); from 7 to 10 April the long end sold off (+45bp on the 30Y between the 4 and 10 April closes) while tariffs were still escalating.

(c) How the improvements change our answers to Q1

Overall. The smooth 60-day patterns of Q1 are mostly macro regime plus clustering, not tariff news. Once these are removed, three findings remain:

1. First-term tariff news had no measurable effect on Treasury yields.
2. In 2025, escalation days, and above all foreign retaliation, pushed *long* yields up and steepened the curve, consistent with a rising term premium (inflation, fiscal and foreign-demand concerns) rather than a flight to safety.
3. De-escalations caused a risk-on rise in front-end yields.

Remaining limitations.

- Second-term groups are small (3–15 events) and concentrated in April 2025, when other forces also hit the long end (deleveraging of Treasury basis trades, swap-spread moves), so the 2025 result is suggestive rather than conclusive.
- A cleaner design would use intraday Treasury futures in 30-minute windows around each announcement and zero-coupon rather than par yields.
- With the relevant curves, foreign yields could measure the reaction in the targeted countries directly.

	Assignment design (Q1)	Improved design (Q2)
Q1.1	Yields fall 10–14bp after Trump actions, mostly at the front end: flight to quality.	**Not robust.** No significant announcement effect (all $p > 0.12$ across the five robustness variants). The 30-day decline disappears (+2.9bp on the 2Y, n.s.) once the baseline is the same-regime drift: it was a rate-cycle effect.
Q1.2	Bull steepening (level down, slope +4.5bp).	Level effect not significant. The **steepening survives** in the short window (slope $+1.6\text{bp}^{**}$ over $[0, +5]$), driven by the second term ($+4.0\text{bp}^{**}$), and comes from the long end *rising*, not from the front end falling.
Q1.3	Little reaction after foreign responses.	**Significant rise** in long yields: 10Y $+3.0\text{bp}^{**}$ and 30Y $+3.2\text{bp}^{**}$ over $[0, +5]$, entirely due to 2025 (10Y +7.1, 30Y +9.5bp).
Q1.4	2nd term: bigger front-end rally.	The front-end rally is not significant at short horizons. The robust difference is at the long end: **first term no reaction, second term yields up and curve steeper**. Treasuries did not act as a safe haven in 2025.
New	—	**De-escalations** raise yields ($+5.1\text{bp}^{**}$ on the 2Y over $[-1, +1]$; e.g. +20bp on 9 April 2025): "risk-on", the mirror image of the flight-to-quality channel, which averaging both types had hidden.

Table 13: Conclusions before and after the methodological improvements.